

mrpSteering

1 Executive Summary

The intend of this module is to implement an MRP attitude steering law with a maximum angular rate of the spacecraft with respect to the reference frame. The module determines and outputs the control torque vector on the spacecraft.

2 Message Connection Descriptions

The following table lists all the module input and output messages. The module msg connection is set by the user from python. The msg type contains a link to the message structure definition, while the description provides information on what this message is used for.

Table 1: Module I/O Messages

Msg Variable Name	Msg Type	Description
guidInMsg	AttGuidMsgPayload	attitude guidance input message
vehConfigInMsg	VehicleConfigMsgPayload	vehicle configuration input message
rwSpeedsInMsg	RWSpeedMsgPayload	(optional) RW speed input message
rwAvailInMsg	RWAvailabilityMsgPayload	(optional) RW availability input message
rwParamsInMsg	RWArrayConfigMsgPayload	(optional) RW configuration parameter input message
cmdTorqueOutMsg	CmdTorqueBodyMsgPayload	commanded torque output message

3 Module Parameters

The following table lists all the module parameters than can be set. The parameters are optional unless indicated (if not specified default is used).

Table 2: Module Parameters

Parameter Name	Type	Units	Default	Description	Bounds
K1	float	[rad/s]	0	Proportional gain applied to MRP errors	Must not be negative (checked in setter)
K3	float	[rad/s]	0	Cubic gain applied to MRP errors	Must not be negative (checked in setter)
omegaMax (required)	float	[rad/s]	0	Maximum rate command of steering control	Must be greater than zero (checked in setter)
ignoreOuterLoopFeedforward	bool	[-]	false	Indicates if feedforward term should be included	N/A
P	float	[N m s]	0	Rate error feedback gain	Must not be negative (checked in setter)

Parameter Name	Type	Units	Default	Description	Bounds
Ki	float	[N m]	0	Integral feedback gain	Must not be negative (checked in setter). If 0, no integral feedback is applied and the corresponding computation is skipped
integralLimit	float	[rad]	0	Limit for integral feedback term (term will be capped by integralLimit)	Must not be negative (checked in setter)
knownTorquePntB_B	Eigen::Vector3f	[N m]	[0, 0, 0]	Known external torque in body frame components	None
controlPeriod	float	[s]	0	control period (1/fsw_rate)	Must be greater than 0

4 Module Assumptions and Limitations

This control assumes the spacecraft is rigid and that the control gains of the attitude control (P, Ki) are chosen such that the decay time is much faster than that of the body rate computation (K1, K3).

5 Initialization

The module is configured by:

```

module = mrpSteering.MrpSteering()
module.modelTag = "mrpSteering"
module.K1 = K1
module.K3 = K3
module.omegaMax = omega_max
module.P = P
module.Ki = Ki
module.integralLimit = integral_limit
module.knownTorquePntB_B = known_torque

```

If the feed-forward term should be ignored:

```

module.ignoreOuterLoopFeedforward = True

```

6 Detailed Module Description

The following text describes the mathematics behind the `mrpSteering` module. Further information can also be found in the journal paper Speed-Constrained Three-Axes Attitude Control Using Kinematic Steering.

6.1 Overview

This module computes a commanded control torque vector $\mathbf{L}_r$ using a rate based steering law that drives a body frame $\mathcal{B} : \{\hat{\mathbf{B}}_1, \hat{\mathbf{B}}_2, \hat{\mathbf{B}}_3\}$ towards a time varying reference frame $\mathcal{R} : \{\hat{\mathbf{R}}_1, \hat{\mathbf{R}}_2, \hat{\mathbf{R}}_3\}$, based on a desired reference frame $\mathcal{B}^* : \{\hat{\mathbf{B}}_1, \hat{\mathbf{B}}_2, \hat{\mathbf{B}}_3\}$ (the desired body frame from the kinematic steering law).

The output message is a body-frame control torque vector. The required attitude guidance message contains both attitude tracking error rates as well as reference frame rates. This message is read in with every update cycle. The vehicle configuration message is only read in on reset and contains the spacecraft inertia tensor about the vehicle's center of mass location. The commanded body rates are read in from the steering module output message.

The reaction wheel (RW) configuration message is optional. If the message name is specified, then the RW message is read in, otherwise the inertia of the wheels is not considered. If the optional RW availability message is present, then the control will only use the RWs that are marked available.

The servo rate feedback control can compensate for Reaction Wheel (RW) gyroscopic effects as well. This is an optional input message where the RW configuration array message contains the RW spin axis $\hat{g}_{s,i}$ information and the RW polar inertia about the spin axis $I_{W,i}$. This is only read in on reset. The RW speed message contains the RW speed Ω_i and is read in every time step. The optional RW availability message can be used to include or not include RWs in the MRP feedback. This allows the module to selectively turn off some RWs. The default is that all RWs are operational and are included.

6.2 Steering Law Goals

The goal of MRP Steering is to drive a body frame $\mathcal{B} : \{\hat{\mathbf{B}}_1, \hat{\mathbf{B}}_2, \hat{\mathbf{B}}_3\}$ towards a time varying reference frame $\mathcal{R} : \{\hat{\mathbf{R}}_1, \hat{\mathbf{R}}_2, \hat{\mathbf{R}}_3\}$. The inertial frame is given by $\mathcal{N} : \{\hat{\mathbf{N}}_1, \hat{\mathbf{N}}_2, \hat{\mathbf{N}}_3\}$. The RW coordinate frame is given by $\mathcal{W} : \{\hat{\mathbf{g}}_{s,i}, \hat{\mathbf{g}}_{t,i}, \hat{\mathbf{g}}_{g,i}\}$. Using MRPs, the overall control goal is

$$\sigma_{\mathcal{B}/\mathcal{R}} \rightarrow 0$$

The attitude error $\sigma_{\mathcal{B}/\mathcal{R}}$ is provided by an upstream module such as `attTrackingError`. The reference frame orientation $\sigma_{\mathcal{R}/\mathcal{N}}$, angular velocity $\omega_{\mathcal{R}/\mathcal{N}}$ and inertial angular acceleration $\dot{\omega}_{\mathcal{R}/\mathcal{N}}$ are assumed to be known.

6.3 MRP Steering Law

To create a kinematic steering law, let $\mathcal{B}^*$ be the desired body orientation, and $\omega_{\mathcal{B}^*/\mathcal{R}}$ be the desired angular velocity vector of this body orientation relative to the reference frame $\mathcal{R}$. The steering law requires an algorithm for the desired body rates $\omega_{\mathcal{B}^*/\mathcal{R}}$ relative to the reference frame.

The desired body rate is computed by

$${}^B\omega_{\mathcal{B}^*/\mathcal{R}} = -\mathbf{f}(\sigma_{\mathcal{B}/\mathcal{R}})$$

where $\mathbf{f}(\sigma)$ is an even function

$$f(\sigma_i) = \arctan \left((K_1\sigma_i + K_3\sigma_i^3) \frac{\pi}{2\omega_{\max}} \right) \frac{2\omega_{\max}}{\pi}$$

and with

$$\sigma_{\mathcal{B}/\mathcal{R}} = (\sigma_1, \sigma_2, \sigma_3)^T$$

and

$$\mathbf{f}(\sigma_{\mathcal{B}/\mathcal{R}}) = \begin{bmatrix} f(\sigma_1) \\ f(\sigma_2) \\ f(\sigma_3) \end{bmatrix}$$

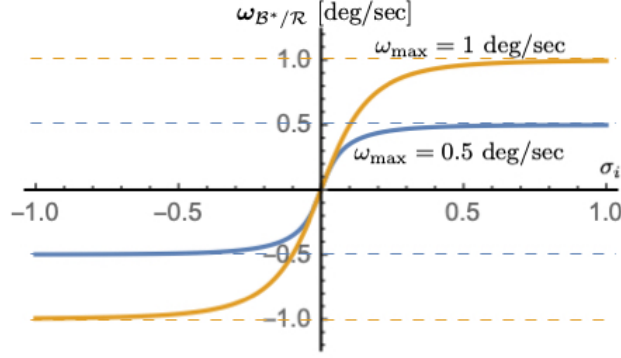


Figure 1: $\omega_{\max}$ dependency with $K_1 = 0.1$, $K_3 = 1$

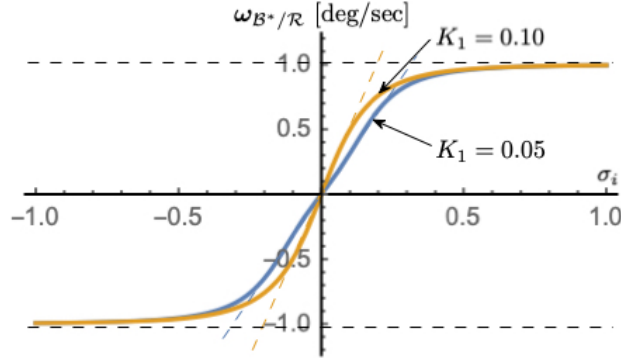


Figure 2: K_1 dependency with $\omega_{\max} = 1$ deg/s, $K_3 = 1$

The required velocity servo loop design is aided by knowing the body-frame derivative of ${}^B\omega_{B^*}/\mathcal{R}$ to implement a feed-forward component. Using the $f()$ function definition, this requires the time derivatives of $f(\sigma_i)$.

$$\frac{{}^B d({}^B\omega_{B^*}/\mathcal{R})}{dt} = \omega'_{B^*}/\mathcal{R} = -\frac{\partial f}{\partial \sigma_{B^*}/\mathcal{R}} \dot{\sigma}_{B^*}/\mathcal{R} = -\begin{bmatrix} \frac{\partial f}{\partial \sigma_1} \dot{\sigma}_1 \\ \frac{\partial f}{\partial \sigma_2} \dot{\sigma}_2 \\ \frac{\partial f}{\partial \sigma_3} \dot{\sigma}_3 \end{bmatrix}$$

where

$$\dot{\sigma}_{B^*}/\mathcal{R} = \begin{bmatrix} \dot{\sigma}_1 \\ \dot{\sigma}_2 \\ \dot{\sigma}_3 \end{bmatrix} = \frac{1}{4} [B(\sigma_{B^*}/\mathcal{R})] {}^B\omega_{B^*}/\mathcal{R}$$

Using the $f()$ definition, its sensitivity with respect to σ_i is

$$\frac{\partial f}{\partial \sigma_i} = \frac{(K_1 + 3K_3\sigma_i^2)}{1 + (K_1\sigma_i + K_3\sigma_i^3)^2 \left(\frac{\pi}{2\omega_{\max}}\right)^2}$$

If `ignoreOuterLoopFeedforward == true`, then $\omega'_{B^*}/\mathcal{R} = 0$ and the computation of the feed-forward component is skipped.

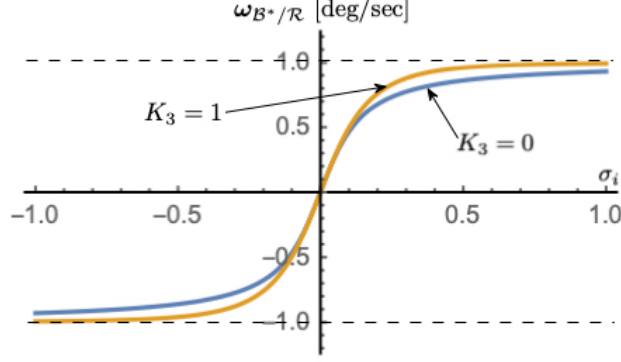


Figure 3: K_3 dependency with $\omega_{\max} = 1$ deg/s, $K_1 = 0.1$

Figures 1-3 illustrate how the parameters $\omega_{\max}$, K_1 and K_3 impact the steering law behavior. The maximum steering law rate commands are easily set through the $\omega_{\max}$ parameters. The gain K_1 controls the linear stiffness when the attitude errors have become small, while K_3 controls how rapidly the steering law approaches the speed command limit.

6.4 Rotational Dynamics

The rotational equations of motion of a rigid spacecraft with N Reaction Wheels (RWs) attached are given by

$$[I_{RW}]\dot{\omega} = -[\tilde{\omega}]([I_{RW}]\omega + [G_s]\mathbf{h}_s) - [G_s]\mathbf{u}_s + \mathbf{L}$$

where the inertia tensor $[I_{RW}]$ is defined as

$$[I_{RW}] = [I_s] + \sum_{i=1}^N (J_{t_i} \hat{\mathbf{g}}_{t_i} \hat{\mathbf{g}}_{t_i}^T + J_{g_i} \hat{\mathbf{g}}_{g_i} \hat{\mathbf{g}}_{g_i}^T)$$

The spacecraft inertia without the N RWs is $[I_s]$, while J_{s_i} , J_{t_i} and J_{g_i} are the RW inertias about the body fixed RW axis $\hat{\mathbf{g}}_{s_i}$ (RW spin axis), $\hat{\mathbf{g}}_{t_i}$ and $\hat{\mathbf{g}}_{g_i}$. The $3 \times N$ projection matrix $[G_s]$ is then defined as

$$[G_s] = [\cdots \hat{\mathbf{g}}_{s_i} \cdots]$$

The RW inertial angular momentum vector $\mathbf{h}_s$ is defined as

$$h_{s_i} = J_{s_i}(\omega_{s_i} + \Omega_i)$$

Here Ω_i is the i^{th} RW spin relative to the spacecraft, and the body angular velocity is written in terms of body and RW frame components as

$$\omega = \omega_1 \hat{\mathbf{b}}_1 + \omega_2 \hat{\mathbf{b}}_2 + \omega_3 \hat{\mathbf{b}}_3 = \omega_{s_i} \hat{\mathbf{g}}_{s_i} + \omega_{t_i} \hat{\mathbf{g}}_{t_i} + \omega_{g_i} \hat{\mathbf{g}}_{g_i}$$

6.5 Angular Velocity Servo Sub-System

To implement the kinematic steering control, a servo sub-system is included which will produce the required torques to make the actual body rates track the desired body rates. The angular velocity tracking error vector is defined as

$$\delta\omega = \omega_{\mathcal{B}/\mathcal{B}^*} = \omega_{\mathcal{B}/\mathcal{N}} - \omega_{\mathcal{B}^*/\mathcal{N}}$$

where the $\mathcal{B}^*$ frame is the desired body frame from the kinematic steering law. Note that

$$\omega_{\mathcal{B}^*/\mathcal{N}} = \omega_{\mathcal{B}^*/\mathcal{R}} + \omega_{\mathcal{R}/\mathcal{N}}$$

where $\omega_{\mathcal{R}/\mathcal{N}}$ is obtained from the attitude navigation solution, and $\omega_{\mathcal{B}^*/\mathcal{R}}$ is the kinematic steering rate command. To create a rate-servo system that is robust to unmodelled torque biases, the state $\mathbf{z}$ is defined as:

$$\mathbf{z} = \int_{t_0}^{t_f} \delta\omega dt$$

Let $[P]^T = [P]$ be a symmetric positive definite rate feedback gain matrix. The servo rate feedback control is defined as

$$\begin{aligned} [G_s]\mathbf{u}_s = & [P]\delta\omega + [K_I]\mathbf{z} - [\tilde{\omega}_{\mathcal{B}^*/\mathcal{N}}] ([I_{\text{RW}}]\omega_{\mathcal{B}/\mathcal{N}} + [G_s]\mathbf{h}_s) \\ & - [I_{\text{RW}}](\omega'_{\mathcal{B}^*/\mathcal{R}} + \dot{\omega}_{\mathcal{R}/\mathcal{N}} - \omega_{\mathcal{B}/\mathcal{N}} \times \omega_{\mathcal{R}/\mathcal{N}}) + \mathbf{L} \end{aligned}$$

Defining the right-hand-side as $-\mathbf{L}_r$, this is rewritten in compact form as

$$[G_s]\mathbf{u}_s = -\mathbf{L}_r$$

The `mrpSteering` module writes the control torque $\mathbf{L}_r$ to the `CmdTorqueBodyMsgPayload` output message.

Downstream, the `rwMotorTorque` module can then be used to compute the array of RW motor torques by solving the typical minimum norm inverse:

$$\mathbf{u}_s = [G_s]^T ([G_s][G_s]^T)^{-1} (-\mathbf{L}_r)$$

7 Additional Information

The value of `integralLimit`, used to limit the degree of integrator windup and reduce the chance of controller saturation. The integrator is required to maintain asymptotic tracking in the presence of an external disturbing torque. For this module, the integration limit is applied to each element of the integrated error vector $\mathbf{z}$, and any elements greater than the limit are set to the limit instead.